Practice with Differentiation So Far . . .

1. If $f(x) = 3x^2 - 5$, what is the equation for the tangent line to the curve when $x = 1$?

(A) $y + 2 = 6(x - 1)$

(B) $y - 2 = -12(x - 1)$

(C) $y - 2 = 6(x - 1)$

(D) $y - 2 = 6(x + 1)$

(E) $y = 6x$

2. If $f(x) = \dfrac{2 + \sin x}{2 - \sin x}$, then $\dfrac{dy}{dx} =$

(A) -1

(B) $\dfrac{4\cos x}{(2 - \sin x)^2}$

(C) $\dfrac{2 + \cos x}{2 - \cos x}$

(D) $\dfrac{2\cos x}{2 - \sin x}$

(E) $\dfrac{(2 + \sin x)^2}{2 - \cos x}$

3. If $f(x) = \dfrac{2x^2}{x + 5}$, then $f'(5) =$

(A) $\frac{1}{2}$

(B) 1

(C) $\frac{3}{2}$

(D) 2

(E) $\frac{5}{2}$

4. $\dfrac{d}{dx}(-5\sec x) =$

(A) $-5\csc x \tan x$

(B) $-5\sec x \tan x$

(C) $5\csc x \tan x$

(D) $5\sec x \cot x$

(E) $5\sec x \tan x$

5. If $f(x) = 4\cot x + 3\tan x$, then $f'(\frac{\pi}{4}) =$

(A) -2

(B) 2

(C) -6

(D) 6

(E) 8

6. $\frac{d}{dx}\left(\frac{1}{4x+2}\right) =$

(A) $\frac{-4}{(4x+2)^2}$

(B) $\frac{-4}{4x+2}$

(C) $\frac{1}{(4x+2)^2}$

(D) $\frac{4}{4x+2}$

(E) $\frac{4}{(4x+2)^2}$

7. If $f(x) = (x^2 + 2x)\sin x$, find $f'(x)$.

(A) $(x^2 + 2x)\sin x + (2x + 2)\cos x$

(B) $x^2 \cos x + 2(\cos x + \sin x) + 2x \sin x$

(C) $x^2 \cos x + 2x(\cos x + \sin x) + 2\sin x$

(D) $x^2 \sin x + 2(\cos x + \sin x) + 2x \sin x$

(E) $x^2 \sin x + 2(\cos x + \sin x) + 2x \cos x$

8. If $f(x) = 3x^3 - 5x + 9$, which of the following is the equation for a tangent line to the curve with a slope $m = 4$?

(A) $y + 7 = 4(x - 1)$

(B) $y + 7 = 4(x + 1)$

(C) $y - 11 = 4(x - 1)$

(D) $y - 11 = 4(x + 1)$

(E) $y - 7 = 4(x + 1)$

9. If $f(x)=\frac{x-6}{2x}$, find the average rate of change of $f'(x)$ on the closed interval $\left[\frac{1}{2},1\right]$.

(A) -18

(B) 84

(C) 18

(D) 100

(E) 64

10. $\frac{d}{dx}\left(\frac{3x-5}{4x+1}\right)=$

(A) $\frac{8}{4x+1}$

(B) $\frac{12x}{4x+1}$

(C) $\frac{1}{(4x+1)^2}$

(D) $\frac{23}{(4x+1)^2}$

(E) $\frac{12x}{(4x+1)^2}$

11. If $f(x)=\frac{\sin x}{x^3}$, then $f'(\pi)=$

(A) $\frac{-3}{\pi^3}$

(B) $\frac{3}{\pi^4}$

(C) $\frac{1}{\pi^4}$

(D) $\frac{1}{\pi^3}$

(E) $\frac{-1}{\pi^3}$

12. If $y=-6x^{-6}+3\sqrt{x}$, then $\frac{dy}{dx}=$

(A) $36x^{-5}+\frac{3}{2\sqrt{x}}$

(B) $36x^{-7}+\frac{3}{2\sqrt{x}}$

(C) $-36x^{-7}+\frac{3}{2\sqrt{x}}$

(D) $\frac{6}{5}x^{-5}+2x^{3/2}$

(E) $-\frac{6}{5}x^{-5}+2x^{3/2}$

13. If $f(x)=\dfrac{2x^3-x^2+3}{x+4}$, find $f'(x)$.

(A) $\dfrac{4x^3+23x^2-8x-3}{x+4}$

(B) $4x^3+23x^2-8x-3$

(C) $\dfrac{x^3+23x^2-8x-3}{x+4}$

(D) $\dfrac{4x^3+23x^2-8x-3}{(x+4)^2}$

(E) $\dfrac{x^3+23x^2-8x-3}{(x+4)^2}$

14. Find the equation of the line normal to the curve $f(x) = \cot(x)$ at $x = \frac{\pi}{4}$.

(A) $y-1=\frac{1}{2}(x-\frac{\pi}{4})$

(B) $y-1=-\frac{1}{2}(x-\frac{\pi}{4})$

(C) $y-1=-\frac{\sqrt{2}}{2}(x-\frac{\pi}{4})$

(D) $y-1=2(x-\frac{\pi}{4})$

(E) $y+1=\frac{1}{2}(x-\frac{\pi}{4})$

15. If $f(x)=\dfrac{1}{1-x^2}$ and $g(x)=\dfrac{f'(x)}{f(x)}$, find $g(x)$

(A) $-\dfrac{2x}{1-x}$

(B) $\dfrac{2x}{1-x^2}$

(C) $\dfrac{x}{1-x^2}$

(D) $-\dfrac{x}{1-x^2}$

(E) $\dfrac{-x}{1-x^2}$

16. If $f(x)=\dfrac{5}{x^4}$, then $f'(2)=$

(A) $\frac{-5}{2}$

(B) $\frac{-5}{8}$

(C) $\frac{-5}{16}$

(D) $\frac{5}{16}$

(E) $\frac{5}{8}$

17. If $f(x) = 5x^2 - 3$, what is the equation for the normal line to the curve when $x = -1$?

(A) $y = 10(x + 1) + 2$

(B) $y = 10(x - 2) + 1$

(C) $y = -10(x + 1) + 2$

(D) $y = \frac{1}{10}(x + 1) + 2$

(E) $y = -\frac{1}{10}(x + 1) + 2$

18. If $f(x) = \frac{x+5}{x-5}$ then $\frac{dy}{dx} =$

(A) $\frac{-10}{x^2 - 10x + 25}$

(B) $\frac{20}{x^2 - 10x + 25}$

(C) $\frac{-10}{x^2 + 10x + 25}$

(D) $\frac{-10}{x-5}$

(E) $\frac{-10}{x-5}$

19. If $f(x) = 5x^{5/2}$, $f'(2) =$

(A) $\sqrt{2}$

(B) 10

(C) $\frac{40}{3}$

(D) $25\sqrt{2}$

(E) 1

20. The equation of the normal line to the curve $y = 3x^2 - 2x$ at $x = 1$ is

(A) $y = \frac{x}{4} + \frac{5}{4}$

(B) $y = \frac{x}{4} + \frac{3}{4}$

(C) $y = -\frac{x}{4} + \frac{5}{4}$

(D) $y = 4x - 3$

(E) None of the above

21. $\frac{d}{dx}(\csc x) =$

(A) $-\csc x \cot x$

(B) $-\csc x \tan x$

(C) $\csc x \cot x$

(D) $-\csc x \tan x$

(E) $\sec x \cot x$

22. The equation of the tangent to the curve $y = x\sin x$ at the point $(\frac{\pi}{2}, \frac{\pi}{2})$ is

(A) $y = x$

(B) $y = x - \pi$

(C) $y = \frac{\pi}{2}$

(D) $y = \pi - x$

(E) $y = x + \frac{\pi}{2}$

23. What is the equation to the normal to the graph of $y = \dfrac{x+2}{x-2}$ at $x = 1$?

(A) $y = \frac{1}{4}x - \frac{13}{4}$

(B) $y = \frac{1}{4}x - \frac{1}{4}$

(C) $y = \frac{1}{4}x - 1$

(D) $y = \frac{1}{4}x - 3$

(E) $y = \frac{1}{4}x - \frac{11}{4}$

24. $\dfrac{d}{dx}\left(\dfrac{\sin x}{x\cos x}\right) =$

(A) $\tan x$

(B) $\sec x$

(C) $\dfrac{x - \sin x \cos x}{x^2 \cos^2 x}$

(D) $\dfrac{x + \sin x \cos x}{x^2 \cos^2 x}$

(E) None of the above

25. $\dfrac{d}{dx}[3x\sin x] =$

(A) $3\cos x$

(B) $3\sin x$

(C) $3x\cos x + 3\sin x$

(D) $3x\sin x + 3\cos x$

(E) $3x\cos x + 3x\sin x$

26. $\frac{d}{dx}[\sin x \tan x] =$

(A) $\sec x + \sin x$

(B) $\sin x + \cos x$

(C) $\tan x + \cos x$

(D) $\tan x + \sec x$

(E) $\sin x + \sec x \tan x$

27. If $y = \tan x \sec x$, $\frac{dy}{dx} =$

(A) $\sec x$

(B) $\sec^3 x$

(C) $\tan x \sec^2 x$

(D) $\sec^3 x + \tan^2 x \sec x$

(E) $\sec^3 x + \tan x \sec^2 x$

28. Base your answer to the following question on the following table.

x	f	f'	g	g'
0	1	2	10	–6
1	2	3	3	–4
2	4	5	1	–2
3	9	10	0	0

If $v(x) = f(x)g(x)$, then $v'(2) =$

(A) –8

(B) 2

(C) –3

(D) 10

(E) 5

29. $\frac{d}{dx}\left(\frac{3}{x} + \frac{5}{x^2} - \frac{7}{x^3}\right)$

(A) $3 + \frac{5}{x} - \frac{7}{x^2}$

(B) $3x + 5x^2 - 7x^3$

(C) $\frac{-3}{x^2} - \frac{10}{x^3} + \frac{21}{x^4}$

(D) $\frac{3}{x^2} + \frac{5}{x^3} - \frac{7}{x^4}$

(E) $\frac{3}{x^2} + \frac{10}{x^3} - \frac{21}{x^4}$

30. $\frac{d}{dx}\left(\frac{x}{6}-\frac{6}{x}\right)$

(A) $\frac{-35}{6}$

(B) $6-\frac{6}{x^2}$

(C) $\frac{1}{6}-\frac{6}{x^2}$

(D) $\frac{1}{6}+\frac{6}{x^2}$

(E) $\frac{x^2}{6}+\frac{6}{x^2}$

31. If $f(x)=(4x^5-3)(3x^2+9)$, then $f'(1)=$

(A) 246

(B) 247

(C) 250

(D) 252

(E) 258

32. If $f(x)=x\sec x$, $f'(\pi)=$

(A) $-\pi$

(B) -1

(C) 0

(D) 1

(E) π

33. $\frac{d}{dx}\left(6x^{2/3}-8x^{1/2}-4\right)=$

(A) $4x^{-1/3}-4x^{-1/2}$

(B) $4x^{1/3}-4x^{-1/2}$

(C) $6x^{-1/3}-4x^{-1/2}$

(D) $\frac{18}{5}x^{5/3}-16x^{3/2}$

(E) $\frac{4}{x^{1/3}}-\frac{4}{x^{1/2}}-4$

34. If $f(x) = 6\sqrt{x} + 2x^{-2}$, then $f'(4) =$

(A) 0

(B) 1

(C) $\frac{5}{4}$

(D) $\frac{23}{16}$

(E) 4

35. If $f(x) = x\cot x$, $f'(\frac{\pi}{2}) =$

(A) 1

(B) -1

(C) $\frac{\pi}{2}$

(D) $-\frac{\pi}{2}$

(E) Zero